Youngwoo Shin

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1. Core Competencies

- **Model-Driven Analysis and Optimization:** Capable of diagnosing internal model mechanisms and enhancing robustness through representation-level analysis, spectral characteristics, and information-theoretic perspectives.
- **Multimodal & Generative AI Expertise:** Broad research spectrum spanning from implementing MLLMs for complex modalities like 4D LiDAR, to analyzing temporal reasoning in VideoLLMs, to establishing effective guidance methods for generative models such as Visual Autoregressive (VAR).
- **End-to-End Research Execution:** Demonstrated ability to lead the full research lifecycle, including problem definition, theoretical hypothesis formulation, dataset/benchmark construction, and model implementation/validation, yielding results in top-tier conferences.

2. Project Experience

[Research] Formulation of Principled Guidance, SSG, for Optimizing Visual Autoregressive Generation

SSG: Scaled Spatial Guidance for Multi-Scale Visual Autoregressive Generation

Accepted to ICLR 2026 (1st Author) — *May 2025 – Feb 2026*

1) Background & Problem Definition

- Identified the train-inference discrepancy in the next-scale generation of Visual Autoregressive (VAR) models, where limited capacity causes models to process redundant low-frequency information instead of generating new, high-frequency content during inference.
- Validated the potential of a training-free guidance technique to control the generative process and optimize performance at inference time, utilizing frequency domain analysis based on an information-theoretic perspective.

2) Key Roles & Implementation

- **Signal Processing-based Analysis:** Diagnosed inefficiencies in the generative information flow via frequency analysis in both latent and pixel spaces.
- **Theoretical Hypothesis & Algorithm Development:** Formulated the SSG algorithm based on Information Bottleneck (IB) theory, designed to minimize information redundancy from previous scales while maximizing scale-specific information gain at the current scale.
- **Minimizing Multi-scale Information Loss:** Proposed Discrete Spatial Enhancement (DSE), a methodology utilizing frequency domain transformation to minimize information loss during multi-scale interpolation, thereby ensuring the robust application of SSG in VAR generation.
- **Validation & Scalability:** Extended beyond VAR to VAR-structured T2I models (HART, Infinity), achieving improved FID/IS and demonstrating validity through spectral analysis.

3) Key Achievements

- **International Conference Acceptance:** Paper accepted to **ICLR 2026** (1st Author), recognized for its theoretical grounding, versatility, and efficiency.
- **Proof of Versatility:** Presented a training-free, efficient optimization solution immediately applicable to various VAR architectures, independent of specific tokenization designs or conditioning modalities, with negligible overhead.

[Research] Diagnosis of Temporal Signal Decay and Training-Free Reinforcement through Interpretability Analysis in VideoLLMs
Before It Fades: Reinforcing Temporal Representations at Inference Time in VideoLLMs
NeurIPS 2026 Under Review (1st Author) — *Mar 2026 – Present*

1) Background & Problem Definition

- VideoLLMs often produce identical temporal predictions even under frame-order reversal, failing temporal reasoning; prior work treated this as an inherent limitation, yet temporal signals are present at intermediate layers but fade before reaching the output.
- Identified the layer region where temporal cues are maximized, presenting a natural intervention point for a representation-level method grounded in the model's temporal divergence profile.

2) Key Roles & Implementation

- **Temporal Divergence Analysis:** Defined the temporal divergence vector τ_l in representation space and its layer-wise profile, revealing a consistent peak-then-decay pattern across diverse VideoLLM architectures including Qwen2.5-VL, Qwen3-VL, and InternVL2.5.
- **Causal Validation:** Confirmed the peak is specific to temporal reasoning via question-conditioned analysis, and established its functional importance for predictions through token-level causal interventions.
- **Temporal Activation Injection (TAI):** Designed a training-free method that extracts temporal signals at the peak layer and reinjects them following the model-specific decay profile, with input-adaptive steering magnitude.
- **Temporal Signal Verification:** Verified that τ_l encodes directional temporal information by reversing the injection sign, which selectively degraded temporal-sensitive categories while leaving invariant ones largely unchanged.
- **Cross-Architecture Validation:** Consistent improvements across 3 VideoLLMs and 4 benchmarks, with gains concentrated on temporal reasoning tasks and negligible impact on non-temporal performance, while remaining orthogonal to training-based methods.

3) Key Achievements

- **International Conference Submission:** Paper under review at **NeurIPS 2026**, proposing a general-purpose diagnostic tool for characterizing temporal information flow in VideoLLMs.
- **Minimal Configuration:** The measured divergence profile determines both the extraction layer and injection schedule, leaving a single scalar as the only free hyperparameter.

[Research] Construction of MLLM Benchmark, B4DL, and Model Development for 4D LiDAR Understanding
B4DL: A Benchmark for 4D LiDAR LLM in Spatio-Temporal Understanding
Accepted to ACM Multimedia 2025 (Co-1st Author) — *Feb 2025 – Oct 2025*

1) Background & Problem Definition

- Increasing importance of 4D LiDAR (3D point cloud + Time) data for understanding dynamic driving environments.
- Need to overcome limitations of existing MLLM research (lack of text datasets corresponding to 4D LiDAR, absence of 4D LiDAR processing architectures) by developing a standardized benchmark and baseline models.

2) Key Roles & Implementation

- **Metatoken Formulation:** Constructed a pipeline to generate LLM-interpretable 4D LiDAR metatokens by textualizing raw metadata, bridging the modality gap by encapsulating spatio-temporal contexts and object interactions.

- **4D LiDAR-Specific MLLM Architecture:** Designed a two-stage alignment methodology that compresses and encodes spatio-temporal 4D LiDAR information to align with LLMs.
- **Validation & Evaluation:** Defined 6 core tasks and conducted quantitative/qualitative evaluations, demonstrating the proposed model's effectiveness in reasoning about dynamic changes over time in 4D scenes.

3) Key Achievements

- **International Conference Acceptance:** Paper accepted to **ACM Multimedia 2025**, recognized for originality and contributions to 4D LiDAR modality interpretation.
- **Setting Industry Standards:** Presented a new direction for 4D LiDAR-MLLM research and released a standard benchmark (dataset & codebase).

[Industry] AI Solution Development & National Project Execution
Vitasoft (AI Engineer / Technical Research Personnel) *Jan 2021 – Jan 2023*

1) KOBACO AiSAC Intelligent Storyboard Generation Service

- **De-identification Pipeline:** Built a pipeline to generate alternative faces or obfuscate identifiable features within reference ad videos to ensure privacy compliance.

2) Patrasche: Object Tracking-based Autonomous Following Cart

- **Person Tracking Algorithm:** Implemented a specific user tracking system using images collected via RGB camera, integrating Person Re-Identification (Re-ID) to identify the target and depth estimation to dynamically adjust following speed of the cart based on distance.
- **Drivable Area Detection:** Developed real-time segmentation methodology to identify flat, drivable surfaces in complex environments including dirt roads, enabling obstacle avoidance and path planning.

3. Education

- **KAIST**, Ph.D. in Electrical Engineering — Mar. 2026 – Present
  Statistical Inference & Information Theory Laboratory (SIIT)
- **KAIST**, M.S. in Electrical Engineering — Mar. 2024 – Feb. 2026
  Statistical Inference & Information Theory Laboratory (SIIT)
- **KAIST**, B.S. in Electrical Engineering — Sep. 2016 – Aug. 2023

4. Industry Experience

- **LG CNS**, Backend Engineer — Jul. 2023 – Feb. 2024
- **Vitasoft**, Computer Vision Researcher — Jan. 2021 – Jan. 2023

5. Skills & Languages

- **Languages:** Python, C, C++
- **Frameworks:** PyTorch, TensorFlow, ONNX, TensorRT
- **Research Areas:** Multimodal Large Language Models (MLLMs), Generative Models, Computer Vision, Model Interpretability, Inference-Time Optimization, 4D Scene Understanding
- **Spoken Languages:** Korean (Native), English (Fluent)